LAB 7:

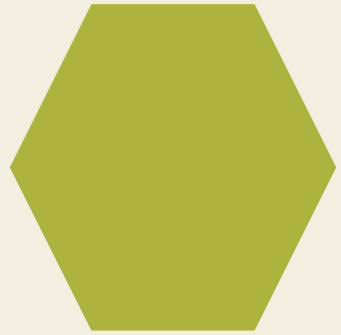
Quantitative Reasoning II

Ibrokhimjon Khamidov

Lab Instructor

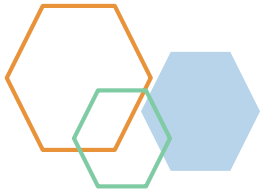
Introduction to Engineering

Fall 2026



TODAY'S LAB · 80 MINUTES

Lab Agenda





Recap of Lecture 7

Limited data, ranges, triangulation, metrics and baselines, honest reporting.

15 min



Instructions for Group Work

The nine-part template (A–I) for deciding with incomplete data.

10 min



Group Work — Decide With Incomplete Data

Diagnose, range, interpret, triangulate, find the flip point, report honestly.

35 min · evaluated

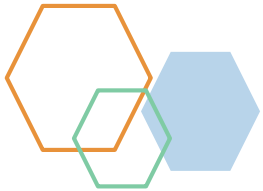


Worked Example: Case #1

LED Lamp Reliability and Energy-Saving Decision, solved step by step.

20 min

Five Ways Data Can Be Incomplete



Problem	What it means	Example
Too little	A few data points only	Three test trips
Missing	No data for the case we care about	No winter test
Noisy	Values scatter around the truth	1.1 to 1.6 kWh/km on the same route
Out of date or unrepresentative	Measured under other conditions	Tests with an empty bus
Wrong definition or unit	Quantity is not what we think	Battery “%”, or “200” with no unit

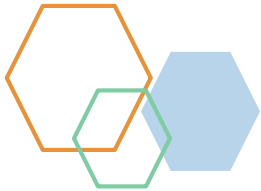
Where numbers come from, and how far to trust them

Measure (high, if the method is stated) · Look up (medium: conditions may differ) · Ask (medium: check it against something) · Estimate (low to medium: state assumptions) · Analogy (low: say how similar it is)

Name the problem first, and write the source and your confidence next to every number.

RECAP · LECTURE 7

Ranges, Small Samples and Triangulation



RANGES

Low, typical, high

Compute the result for each value, not just one. Winter bus: 160 kWh ÷ 1.5 / 1.8 / 2.0 kWh/km = 107 / 89 / 80 km against a 96 km need: only the optimistic case passes.

SMALL SAMPLES

Say how many (n)

Three trips: 1.1, 1.3, 1.6 kWh/km. Mean 1.33, median 1.3. One odd value moves the mean a lot, the median barely. Write n next to every number; do not claim more than n points can show.

TRIANGULATE

Estimate it three ways

Faculty building: 8,000 m² × 0.35 ≈ 2,800 kWh/day; 94 kW × 24 h ≈ 2,260; monthly bill 70,000 ÷ 30 ≈ 2,330. Independent routes that agree build trust; disagreement tells you where to look.

A range tells the reader how far to trust the answer. When more precision cannot change the decision, stop refining.

RECAP · LECTURE 7

Same Data, Different Readings

METRIC

The decision picks the metric

Typical rider: median (≈5 min). Total waiting: average (≈6 min). The unlucky: 90th percentile (≈13 min). Worst case: maximum (≈16 min). Same bus data, different questions.

BASELINE

A percentage of what?

An LED retrofit saves 12 kW: that is 60% of the corridor lighting (12 ÷ 20 kW) but 7% of the building peak (12 ÷ 180 kW). Two true percentages, two stories.

MISLEADING COMPARISONS

Each needs a question

Different denominators: per what? Cherry-picked days: which days were left out? Truncated axis: where does it start?

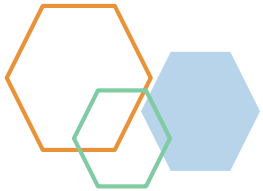
Four questions for any interpretation

1 What decision is this number for? 2 Which metric fits it? 3 Which baseline, boundary and time basis is used? 4 Which assumption, if wrong, would flip the conclusion?

Ask these before you accept a result, and before you present one.

RECAP · LECTURE 7

Reporting Honestly



RESULT

The answer

A plain sentence a reader can use.

RANGE

How far to trust it

Low to high, not one value.

TOP ASSUMPTION

What it rests on

The one that matters most.

WHAT WOULD CHANGE IT

What to learn next

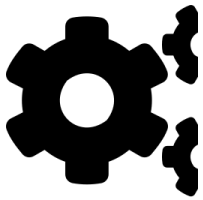
The data that would move it.

Lecture example: can one bus cover a winter day of 96 km?

Result: probably not (89 km against 96 km). Range: 80 to 107 km; only the optimistic case passes. Top assumption: 1.8 kWh/km winter use, low confidence. What would change it: a real winter trip, or a mid-day top-up.

A good report gives the answer, how far to trust it, and what would change it.

GROUP WORK · EVALUATED · 40 MINUTES



Engineering Decisions with Incomplete Data

Different case scenario exercises.

Your Task: Decide With Incomplete Data (1/2)

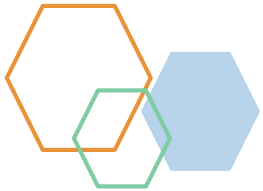
A **Diagnose the data problems.** Find at least 3 issues. Name each: too little, missing, noisy, out of date / unrepresentative, or wrong definition / unit. Say why it is a problem and how to get a better number.

B **Source and confidence.** Pick 4–5 important numbers. For each: value or range, source (measure, look up, ask, estimate, analogy), confidence, and a short reason.

C **Build a low / typical / high range.** Choose one uncertain variable that matters. Recalculate the main result for each value and write: “The output is about __ to __, typical __.”

D **Same data, two interpretations.** Write a favourable and a critical reading. Which numbers support both? What extra data would settle the argument?

Your Task: Decide With Incomplete Data (2/2)



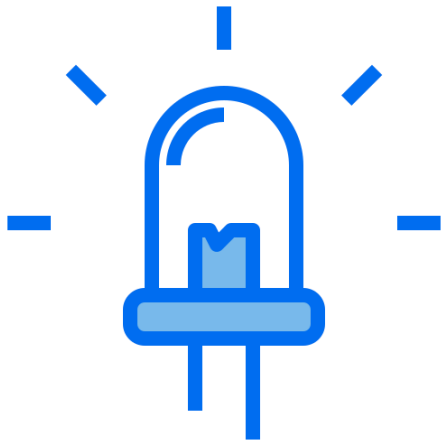
E **Choose the right metric.** For 3 decision questions pick average, median, 90th percentile, maximum, peak or a rate per unit. Say why, and why another metric could mislead.

F **Test the baseline.** Express one result against two different baselines. Show both calculations and explain why both can be true but tell different stories.

G **Triangulate one key quantity.** Estimate it at least 2 independent ways. Do the results agree closely enough to support the decision?

H **What would flip the decision?** Find the most important uncertain assumption and the threshold at which the decision changes. Show the calculation or reasoning.

I **Final honest report.** Result, range, top assumption, what would change it, in one sentence. Then give a 60-second team report-out.



LED Lamp Reliability and Energy-Saving Decision

WORKED EXAMPLE · CASE #1

LED Lamp Reliability and Energy Saving

1 LED Lamp Reliability and Energy-Saving Decision



A manufacturer is considering replacing an older LED lamp model with a new high-efficiency version for a large university lighting contract. The old lamp is rated at 18 W and the new one at 12 W. The supplier claims the new model provides the same light output and lasts 40,000 h, but this lifetime comes from the supplier datasheet rather than the university's own testing.

The engineering team tested only five lamps from one production batch. Their measured powers were 11.7, 12.1, 12.3, 11.9 and 12.6 W. Only two lamps have been operated for more than 2,000 hours, so there is no direct evidence yet about long-term lifetime.

The university currently has approximately 6,000 corridor lamps, but the facilities manager says this number was counted before a new building opened. A technician estimates that the actual total may now be between 6,300 and 6,700 lamps.

Electricity use is reported in two different ways. An older energy report says corridor lighting accounts for 20 kW of the university's night-time electrical load, while a recent facilities estimate says it is closer to 75 kW across all buildings.

The procurement manager says: "The new lamp saves about one-third of the energy, so we should replace everything." A maintenance engineer responds: "We do not know whether these lamps will really last 40,000 hours."

Students must decide how confidently the university can estimate annual energy savings and whether the replacement decision is justified now.

Quantity	Available information
Old lamp power	18 W
New lamp tests	11.7, 12.1, 12.3, 11.9, 12.6 W
Supplier lifetime claim	40,000 h
Number of lamps	6,000 old count; estimated 6,300–6,700 today
Operating time	Estimated 10–14 h/day
Electricity tariff	700–1,000 UZS/kWh depending on contract period

LAB 7: Diagnose data issues • Source & confidence • Low/typical/high • Two interpretations • Choose metric • Check baseline • Triangulate • Flip point • Honest report

Decide how confidently the university can estimate annual energy savings, and whether replacing all lamps is justified now.

SOLUTION · PART A

Diagnose the Data Problems

Quantity / statement	Problem type	Why it is a problem	A better number
Only 5 lamps tested	Too little; unrepresentative	n = 5 from a single batch; only 2 lamps ran over 2,000 h	Test several batches and more lamps
40,000 h lifetime	Missing	Supplier datasheet; no university evidence yet	Accelerated life test, field failure data
6,000 lamps	Out of date	Counted before a new building opened; now 6,300–6,700	Recount from the asset register
20 kW vs 75 kW	Wrong definition / boundary	Different baselines for “corridor lighting”	Ask what each includes; check bottom-up (6,500 × 18 W ≈ 117 kW)
10–14 h/day; 700–1,000 UZS/kWh	Noisy (wide spread)	Estimates, not logs; tariff depends on contract period	Log hours; use the contract tariff

Five issues, and the fix depends on which type each one is.

SOLUTION · PART B

Source and Confidence

Quantity	Value / range	Source	Confidence	Short reason
New-lamp power	11.7–12.6 W; mean 12.1 W (SD 0.35, n = 5)	Measured	Medium	Method stated, but one batch and n = 5
Old-lamp power	18 W	Rated (look up)	Medium	Rating, not measured; check a few old lamps
Lamp count	6,300–6,700 (was 6,000)	Technician estimate	Medium-low	Old count; new building opened
Operating time	10–14 h/day	Estimate (ask)	Low-medium	No logs; ±17% around 12 h
Lifetime	40,000 h (≈ 9 years at 12 h/day)	Supplier datasheet	Low-medium	Not tested locally; 2 lamps over 2,000 h

The power numbers are the strongest; hours and lifetime are the weakest. That decides what we test next.

SOLUTION · PART C

Build a Low / Typical / High Range

Case	Saving per lamp [W]	Lamps [–]	Hours [h/day]	Tariff [UZS/kWh]	Energy saved [MWh/yr]	Cost saved [M UZS/yr]
Low	5.4 (18 – 12.6)	6,300	10	700	124	87
Typical	5.88 (18 – 12.12)	6,500	12	850	167	142
High	6.3 (18 – 11.7)	6,700	14	1,000	216	216

Energy saved = saving per lamp × lamps × hours × 365 ÷ 1,000. Typical: 5.88 [W] × 6,500 × 12 [h/day] × 365 ÷ 1,000 ≈ 167,000 [kWh/yr].

Range statement: about 124 to 216 MWh per year, typical 167 (n = 5 lamps, one batch). Cost: about 87 to 216 million UZS per year.

Which uncertain variable matters most? Change one at a time, others typical

OPERATING HOURS

139 to 195 MWh/yr

10 → 14 h/day

LAMP POWER SAVING

154 to 179 MWh/yr

5.4 → 6.3 W

LAMP COUNT

162 to 173 MWh/yr

6,300 → 6,700

SOLUTION · PART D

Same Data, Two Interpretations

FAVOURABLE · PROCUREMENT MANAGER

“Replace everything.”

All five new lamps measure close to 12 W against 18 W: a saving of about one-third per lamp (30–35%) that is very likely real.

CRITICAL · MAINTENANCE ENGINEER

“We do not know they will last.”

The lifetime is unverified, the lamp count and hours are uncertain, and energy saving alone does not prove a lower lifecycle cost.

NUMBERS THAT SUPPORT BOTH

Measured power and the saving

12 W vs 18 W (≈ 6 W per lamp) is true in both readings. The argument is about life, hours and price, not about the watts.

EXTRA DATA THAT WOULD SETTLE IT

Reliability, hours and price

A multi-batch field pilot with failure logs, logged operating hours, the current lamp count, and the purchase plus installation price.

Both sentences are true. The extra data decides which problem matters more.

SOLUTION · PART E

Choose the Metric for the Decision

Decision question	Best metric	Why, and why another could mislead
How much energy will be saved?	Annual kWh saved (and a range)	Uses the real lamp count and hours; a % per lamp hides how many lamps there are
What does a typical new lamp draw?	Median (with mean, range and n)	With n = 5 the mean and median agree (12.1 W); say n and show the range
Will the lamps last long enough?	Failure rate / survival hours	Mean power says nothing about reliability; one datasheet figure is not evidence
Is replacement worth the money?	Payback or cost per year of service	Energy saving alone ignores the purchase price and early failures

The decision picks the metric, not the other way round.

SOLUTION · PART F

Test the Baseline

Result	Baseline	Calculation	Share	Interpretation
Saving per lamp: 5.88 W	Old lamp: 18 W	5.88 ÷ 18	33%	Procurement's "one-third" is true per lamp (30–35%)
Total saving ≈ 38 kW (6,500 × 5.88 W, all on)	Old connected lighting: 6,500 × 18 W ≈ 117 kW	38.2 ÷ 117	33%	Same boundary, same percentage
Same 38 kW saving	Facilities estimate: 75 kW	38.2 ÷ 75	51%	Too high: 75 kW cannot be all lamps on at once
Same 38 kW saving	Old report: 20 kW	38.2 ÷ 20	191%	Impossible (> 100%): the boundary is not the same

Always ask: a percentage of what? A result above 100% is a clue that the baseline boundary is wrong.

Use one boundary before and after the retrofit, and state it: corridor lamps only, all lighting, or the whole building.

SOLUTION · PART G

Triangulate the Annual Saving

Route	Method / source	Working	Result [MWh/yr]	Confidence
1 Bottom-up (lamps)	Measured power × counted lamps × hours	6,500 × 5.88 W × 12 h × 365 ÷ 1,000	≈ 167	Medium-low
2 Top-down (75 kW)	Facilities estimate × one-third saving	75 kW × 0.327 × 12 h × 365	≈ 107	Low-medium
3 Top-down (20 kW)	Old energy report × one-third saving	20 kW × 0.327 × 12 h × 365	≈ 29	Low

AGREEMENT CHECK

Routes 1 and 2 agree within a factor of 1.6 (same order of size): the saving is real and large, but its exact size is not tight. Route 3 is six times lower; it would mean only about 1,100 lamps are on, so the 20 kW figure is probably a subset. Next step: find out how many lamps are on at night and for how many hours.

SOLUTION · PART H

What Would Flip the Decision?

ENERGY CLAIM

Robust

It flips only if the new lamp really draws close to 18 W. The tests show 11.7–12.6 W, so the energy saving is not in doubt.

REPLACE-NOW DECISION

Depends on life and price

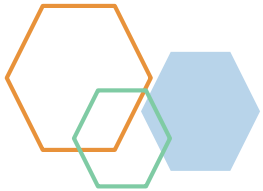
Break-even extra price per lamp ≈ annual saving per lamp × years the lamp really lasts. Typical saving: 25.8 kWh × 850 UZS ≈ 21,900 UZS per lamp per year (range 13,800–32,200).

If the lamp really lasts…	Years at 12 h/day	Break-even extra price per lamp
40,000 h (the supplier claim)	≈ 9.1	≈ 200,000 UZS
20,000 h (half the claim)	≈ 4.6	≈ 100,000 UZS
10,000 h (a quarter of the claim)	≈ 2.3	≈ 50,000 UZS

Lamp price and installation cost are not given, so the threshold is a formula, not one number. Best next measurement: a multi-batch field pilot.

SOLUTION · PART I

Final Honest Engineering Report



RESULT

Saving real, payback unproven

About one-third less energy per lamp, ≈ 167 MWh/yr across the campus.

RANGE

124 to 216 MWh/yr

Top-down check gives ≈ 107. Cost saved 87–216 M UZS/yr.

TOP ASSUMPTION

Hours and real lifetime

12 h/day and 40,000 h are both unverified.

WHAT WOULD CHANGE IT

Pilot, count, price

Logged hours, current lamp count, multi-batch failure data, lamp price.

ONE-SENTENCE REPORT

“The new lamps should save about 170 MWh a year (range about 125 to 215, from 5 lamps of one batch), but because the 40,000 h life and the price are unverified we recommend a one-building pilot before replacing all 6,500 lamps.”

Thank You — Questions?

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